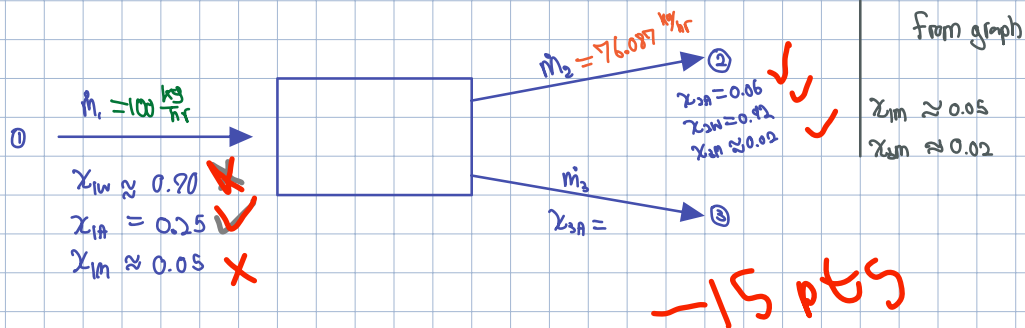


?s : 1, 2, 3, 4

Due November 14th

① (1) Murphy 5.40

P5.40 Acetic acid is to be extracted from water in a single mixer-settler unit using pure methyl isobutyl ketone (MIBK) as the solvent. The feed contains 25 wt% acetic acid in water, and the water-rich product stream must contain no more than 6 wt% acetic acid. Calculate the kg solvent/kg feed and the compositions and flow rates of the two streams leaving the process. (Refer to Fig. 5.15.)



Basis : $\dot{m}_1 = 100 \text{ kg/hr}$

W : $x_{1w} \dot{m}_1 = x_{2w} \dot{m}_2$ *there is also water on $\dot{m}_3$* ✓
 $100 \times 0.9 = 0.92 \dot{m}_2$
 $90 = 0.92 \dot{m}_2$
 $\dot{m}_2 = 97.826 \text{ kg/hr}$ X

A : $x_{1a} \dot{m}_1 = x_{2a} \dot{m}_2 + x_{3a} \dot{m}_3$ ✓
 $0.25 \times 100 = 0.06 \times 97.826 + 1 \times \dot{m}_3$
 $25 = 5.8696 + \dot{m}_3$
 $\dot{m}_3 = 19.1304 \text{ kg/hr}$ X

$\dot{m}_1 = 100 \text{ kg/hr}$
 $\dot{m}_2 = 97.826 \text{ kg/hr}$
 $\dot{m}_3 = 19.1304 \text{ kg/hr}$ X

$x_{1m} \approx 0.05 \frac{\text{kg Solvent}}{\text{kg Feed}}$ X

10/25

(2) Murphy 5.41 The figure included in the problem statement shows c_p in mmol/L plotted on the horizontal axis and q_p in mmol/g plotted on the vertical axis. The data are given at the right. Fit these data to two adsorption models: (1) the Langmuir isotherm:

$$q_p = q_{\infty} \frac{K_L c_p}{1 + K_L c_p}$$

where q_{∞} and K_L are adjustable parameters and (2) the Freundlich isotherm:

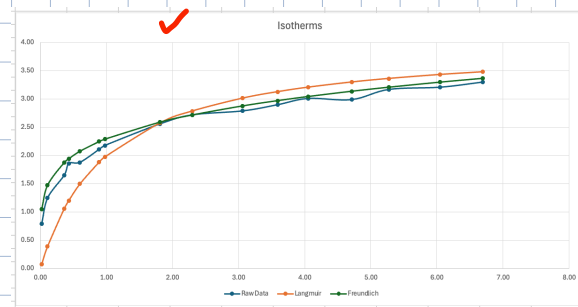
$$q_p = k_F c_p^{1/n}$$

where k_F and n are another set of adjustable parameters. Try fitting the data to both models. Which is better? Use that model to solve the problem.

c_p (mmol/L)	q_p (mmol/g)
0.02	0.79
0.11	1.25
0.36	1.65
0.43	1.86
0.60	1.88
0.89	2.11
0.98	2.18
1.81	2.56
2.30	2.72
3.06	2.79
3.59	2.90
4.05	3.01
4.71	2.99
5.27	3.17
6.05	3.21
6.69	3.30

How did you calculate these coeff

DATA	Langmuir isotherm	Freundlich isotherm
	$Q_p = Q_{\infty} \cdot (K_L \cdot C_p) / (1 + K_L \cdot C_p)$	$Q_p = K_F \cdot C_p^{1/n}$
	Q_{∞} L	n K _F
	0.02 0.79	0.02 0.79
	0.11 1.25	0.11 1.25
	0.36 1.65	0.36 1.65
	0.43 1.86	0.43 1.86
	0.60 1.88	0.60 1.88
	0.89 2.11	0.89 2.11
	0.98 2.18	0.98 2.18
	1.81 2.56	1.81 2.56
	2.30 2.72	2.30 2.72
	3.06 2.79	3.06 2.79
	3.59 2.90	3.59 2.90
	4.05 3.01	4.05 3.01
	4.71 2.99	4.71 2.99
	5.27 3.17	5.27 3.17
	6.05 3.21	6.05 3.21
	6.69 3.30	6.69 3.30



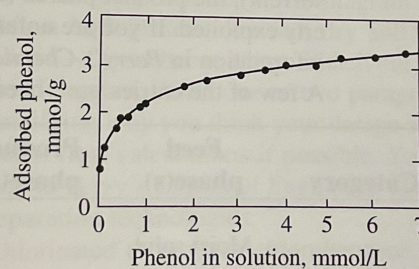
— 5 pts 20/25

Setting

$n = 5$ X

$k_F = 2.3$ X, I found Freundlich more accurate J

P5.41 Trace organic contaminants in waste water readily adsorb onto a solid called *activated carbon*, which is not very different from charcoal briquets. Equilibrium data for adsorption of phenol onto activated carbon is shown. One liter of wastewater containing 1 g phenol is contacted with 5 g activated carbon at 20°C. At equilibrium, what fraction of the phenol is adsorbed?



$$q_p (\text{mmol/g}) = k_f C_p^{(n)}$$

$$q_p = 2.3 C_p^{1/5}$$

$$1 \frac{\text{g Phenol}}{\text{L}} + 5 \text{g Carbon}$$

$$M_{\text{Phenol}} = 94.11 \text{ g/mol} \quad \checkmark$$

$$1 \frac{\text{g Phenol}}{\text{L}} \times \frac{1 \text{ mol}}{94.11 \text{ g}} \times \frac{1000 \text{ mmol}}{1 \text{ mol}} = 10.63 \frac{\text{mmol}}{\text{L}} \quad \checkmark$$

$$q_p = 2.3 C_p^{1/5} \quad \checkmark$$

$$= 2.3 (10.63 \frac{\text{mmol}}{\text{L}})^{1/5} = 3.69 \frac{\text{mmol}}{\text{g}}$$

OK
not the
answer.

(3) Air is heated from 25°C to 150°C prior to entering a combustion furnace. The change in specific enthalpy associated with this transition is 3640 J/mol. The flow rate of air at the heater outlet is 1.25 m³/min and the air pressure at this point is 122 kPa absolute.

- (a) Calculate the heat requirement in kW, assuming ideal gas behavior and that the kinetic and potential energy changes from the heater inlet to the outlet are negligible.
- (b) Would the value of $\Delta \hat{E}_k$ [which was neglected in part (a)] be positive or negative, or would you need more information to answer? If you need more information, what would that information be?

②

$$Q_o = 1.25 \frac{\text{m}^3}{\text{min}} \quad \hat{H} = 3640 \frac{\text{J}}{\text{mol}}$$

$$P_o = 122 \text{ kPa} \quad 67 = 150 - 25 = 125$$

$$\dot{U} + \cancel{\dot{E}_k} + \cancel{\dot{E}_p} = Q + \cancel{W}$$

$$Q = \dot{U} = \dot{n} \hat{H}$$

$$= \dot{n} \times 3640 \frac{\text{J}}{\text{mol}}$$

$$Q = \dot{n} \times 3640 \frac{\text{J}}{\text{mol}}$$

$$P\dot{V} = \dot{n}RT$$

$$\dot{n} = \frac{P\dot{V}}{RT} = \frac{(122 \text{ kPa} \times \frac{1 \text{ bar}}{100 \text{ kPa}}) \times (1.25 \frac{\text{m}^3}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ sec}} \times (\frac{1000 \text{ m}}{1 \text{ m}})^3)}{((150^\circ\text{C} - 25^\circ\text{C}) + 273.15) \times (83.144 \frac{\text{bar} \cdot \text{cm}^3}{\text{mol} \cdot \text{K}})}$$

$$= 0.7678 \frac{\text{mol}}{\text{s}} \quad \downarrow$$

$$Q = (0.7678 \frac{\text{mol}}{\text{s}}) \times 3640 \frac{\text{J}}{\text{mol}} = 2794.75 \frac{\text{J}}{\text{s}}$$

$$= 2.795 \text{ kW} \quad \downarrow$$

③ would need to know more info. Need to know if inlet and outlet have the same cross sectional area. Or what size the openings are. ↓

(4) Murphy 6.26

P6.26 Calculate the energy change, in J, for 1 gmol of H_2O undergoing the following changes:

- (a) accelerating from 0 to 100 km/h
- (b) rising 100 meters
- (c) increasing in pressure at 100°C from 0.1 bar to 1 bar
- (d) increasing in temperature from 0 to 100°C at 5 bar
- (e) dissociating to H_2 and O_2 at 25°C
- (f) undergoing a nuclear reaction where all mass converts to energy ($E = mc^2$)

$$\textcircled{a} \Delta E_k = \frac{1}{2} m (v_2 - v_1)^2$$

$$1 \text{ mol } \text{H}_2\text{O} \times \frac{18.015 \text{ g}}{1 \text{ mol}} \times \frac{1 \text{ kg}}{1000 \text{ g}} = 0.018015 \text{ kg}$$

$$0 \text{ km/h} = 0 \text{ m/s}$$

$$100 \text{ km/h} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hr}}{3600 \text{ s}} = 27.78 \text{ m/s}$$

$$\begin{aligned} \Delta E_k &= \frac{1}{2} \times (0.018015 \text{ kg}) \times (27.78 - 0)^2 \\ &= 6.95 \text{ J} \quad \checkmark \end{aligned}$$

21/25

$$\textcircled{b} \Delta E_p = m g \Delta h$$

$$\begin{aligned} &= (0.018015 \text{ kg}) \times (9.81 \text{ m/s}^2) \times (100 - 0) \\ &= 17.67 \text{ J} \quad \checkmark \end{aligned}$$

$$\textcircled{c} \quad \Delta P = 1 - 0.1 \quad t = 100^\circ\text{C}$$

$$\Delta H = \Delta U + P\hat{V}$$

$$\Delta U = \Delta H - \Delta P\hat{V}$$

$$P = 0.1 \text{ bar} : \begin{aligned} \hat{H} &= 2687.5 \\ \hat{U} &= 2515.5 \\ \hat{V} &= 17.197 \end{aligned}$$

$$\Delta \hat{U} = \begin{aligned} &2506.2 \quad - \quad 2516.5 \\ &\text{@ 1 bar} \quad \quad \text{@ 0.1 bar} \end{aligned}$$

$$P = 1 \text{ bar} : \begin{aligned} \hat{H} &= 2675.8 \\ \hat{U} &= 2506.2 \\ \hat{V} &= 1.6959 \end{aligned}$$

$$= -9.3 \frac{\text{kJ}}{\text{kg}} \times 0.018015 \text{ kg}$$

$$= -1,675.4 \text{ J}$$

done, -2 pts

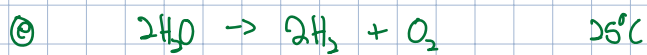
$$\textcircled{d} \quad \Delta T = \frac{100}{5} - \frac{0}{1} \quad \text{at } 5 \text{ bar}$$

$$\Delta \hat{H} = m(c_p \Delta T)$$

$$= (0.018015 \text{ kg}) \times \left(4.19 \frac{\text{J}}{\text{g}^\circ\text{C}} \times \frac{1000 \text{ g}}{1 \text{ kg}} \right) \times (100 - 0)$$

$$= 7,530.27 \text{ J}$$

✓



$$\Delta H = \sum nH_p - \sum nH_r$$

$$= (2H_{H_2} + 1H_{O_2}) - (2 \times 142.9)$$

$$= 0 - 285.8$$

$$= -285.8 \frac{\text{kJ}}{\text{mol}}$$

wrong sign
-2 pts

④

$$E = mc^2$$

$$= (0.018015 \text{ kg}) \times (3 \times 10^8)^2$$

$$= 1.62135 \times 10^{15} \text{ J}$$

$$= 1.62 \times 10^{12} \text{ kJ} \quad \checkmark$$